

ME30: Engineering Thermodynamics

Objective: ME30 is designed to offer a thorough treatment, at the undergraduate level, of both the theory and applications behind thermodynamic systems. Examples of thermodynamic systems with applications to energy conversion and propulsion will be given, and constraints (both fundamental and practical) are discussed. Specific goals of this introductory-level thermodynamic course are:

1. Provide a sound understanding about the fundamentals of thermodynamics.
2. Introduce elementary principles of thermodynamic cycle analysis as applied to energy conversion and representative propulsion systems.
3. Provide a working knowledge of and tools to analyze various propulsion systems and thermodynamic cycles of practical relevance, such as refrigerator cycle, reciprocating engines, turbojets, and rocket engines.

Outcome: By the end of the course, you should:

1. Have a comprehensive understanding about classical thermodynamics, including 1st and 2nd law, state relations, and thermodynamic cycles.
2. Be able to perform thermodynamic cycle analysis, including energy balance and characterization of thermal efficiency, heat, and work, as applied to thermodynamic and energy-conversion systems.
3. Have a basic understanding about environmental challenges and technological issues of advanced thermodynamic and propulsion systems.

Prerequisites: Math 21

Instructor: Matthias Ihme

Office hours: Monday: 4:00 – 5:00 PM and by appointment

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Lectures: 530-127

9:30AM-10:20AM Monday, Wednesday, and Friday

Lectures will *primarily focus* on the principles of thermodynamics and how to apply these principles to engineering systems; we will discuss examples to illustrate concepts, but we will complement the lecture with a *weekly tutorial* to go through example problems and apply the methods and theory learned in lecture.

Tutorials: Weekly tutorials will be held by the TAs. These tutorials complement the lecture material and largely focus on applying concepts for solving problems that are similar to those of the assignments. Therefore, attendance is strongly encouraged. Tutorial problems are posted 2 days before the tutorial session, and solutions will be posted after the tutorial. Please see announcements on canvas for tutorial schedule.

Grading and grading policy:

Grading:	Homework	40%
	Midterm 1	15%
	Midterm 2	15%
	Final Exam	30%

Grading policy: Homework submission via gradescope; weekly homework assignments (except during the week of the midterm). In the grading, we will discard the homework set with the lowest score. So, out of

the 7 homework sets assigned, only 6 homework sets will be considered. **Late work** is accepted with a *loss of 15% per day for a maximum of 3 days* (including weekend). Valid excuses are not penalized but must be discussed (at least 3 days) in advance with the instructor.

Course webpage: <http://canvas.stanford.edu>

CAs: Priyanka Muhunthan (pmuhun@stanford.edu)
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Information about meeting times for all office hours is provided on canvas,
and will be regularly updated

Textbook: Engr30 Textbook “Thermodynamics: An Engineering Approach”, which is a consolidated paperback version of Çengel, Y. A. and Boles, M. A. “Thermodynamics: An Engineering Approach.” Older version are available in online version through the [Hathi Trust via the Stanford library system](#)

Policy and lecture rules:

- All material covered in class may be on a test
- No open laptops during class (tablets for taking notes are allowed)
- No emailing or other electronic/social network media allowed during lecture

Tentative course schedule, reading assignments, and problem sets

Week	Lecture	Date	Topic	Reading ¹	PS Due
1	1	9/23	Overview, introduction	Ch. 1-1,2,3,4	
	2	9/25	Forms of energy, units, problem solving techniques and problem set guidelines	Ch. 1-6,7,8,9;	
	3	9/27	Nomenclature, energy transformation	Ch. 2-1,2,3,4,5	
2	4	9/30	Properties of pure substances and phase changes	Ch. 3	
	5	10/02	Property table and property diagrams		#1
	6	10/04	Equation of state, ideal gas and real fluids		
3	7	10/07	First law, control mass (CM), boundary work	Ch. 2-6,7,8; Ch.-4	
	8	10/09	First law, control mass (CM), specific heat		#2
	9	10/11	First law, control volume (CV)		
4	10	10/14	Steady flow systems	Ch. 5	
	11	10/16	Examples of CM, CV: Otto Cycle, Refrigerator Cycle, nozzle, diffuser, turbine, compressor	Ch. 9-4,5,6	#3
	12	10/18	Examples of CM, CV: Otto Cycle, Refrigerator Cycle, nozzle, diffuser, turbine, compressor	Ch. 11-1,2,3	
5	13	10/21	<i>Midterm #1 (50 min)</i>		
	14	10/23	Concept of entropy	Ch. 6-1; Ch. 7-1,2	
	15	10/25	Entropy and statistical thermodynamics		
6	16	10/28	Entropy as a property	Ch. 7-3,4,7,8,9	
	17	10/30	Entropy balance and generation	Ch. 6-6; Ch. 7-13	#4
	18	11/01	Entropy balance and second law		
7	19	11/04	Carnot cycle	Ch. 6-3,4,7,8,10,11	
	20	11/06	Entropy changes and isentropic processes	Ch. 7-5,12	#5
	21	11/08	Gibbs equation		
8	22	11/11	Reversible steady flow work	Ch. 9-1; Ch. 7-10,11; Ch. 11-1,2,3	
	23	11/13	<i>Midterm #2 (50 min)</i>		
	24	11/15	Isentropic efficiency, entropy balance		
9	25	11/18	Cycle analysis: Vapor power cycle	Ch. 8-1,2,3,4	
	26	11/20	Cycle analysis: Otto cycle	Ch. 9-5	#6
	27	11/22	Cycle analysis: Diesel cycle	Ch. 9-6	
9		11/25	Thanksgiving recess		
		11/27	Thanksgiving recess		
		11/29	Thanksgiving recess		
11	28	12/02	Cycle analysis: Gas turbine and Brayton cycle	Ch. 9-8,9,10,11	
	29	12/04	Alternative energy conversion processes		#7
	30	12/06	Statistical interpretation of thermodynamics		
Final Exam: 12/12, 8:30-11:30AM (tentative)					

Travel: 10/4, 10/7, 10/25

¹ All reading material is in reference to Eng30-textbook or Çengel & Boles, “Thermodynamics, An Engineering Approach” 7th edition

ME30 Problem Set Guidelines

Principle Guidelines: *Two thirds* of what you will learn in this course will be through tutorials and problem sets. A problem set will be assigned each week. Problem sets are due the following week *before the beginning of the class*. Late work is accepted with a *loss of 15% per day for a maximum of 3 days* (including weekend). Valid excuses are not penalized but must be discussed (at least 3 days) in advance with the instructor. Other guidelines:

- Homework submission and grading will be done through gradescope
- Homework solutions must be submitted in neat, professional, and legible form.
- Honor Code² applies to all homework problems, midterms, and final exam:
- Homework sets may include mathematical derivations, for which partial credit is given. Therefore, unless otherwise stated, *symbolic software packages* (such as MAPLE, Mathematica, Matlab-symbolic tool-box, ...) are not allowed for solving the homework problems.

Solution Approach: For many students ME30 is the first exposure to engineering thermosciences. As such, it is important to get off to a good start by writing clear, accurate analyses of the problems you solve. Bear in mind that in evaluating your homework it is not the numerical value that you arrive at that is of interest, but the *process* that you used to arrive at it. For this reason, it is critically important that your solution accurately depicts how you went about solving the problem.

In order to help you in depicting your solution process, follow these guidelines in writing up your solutions for ME30. *Points will be deducted for incomplete or sloppy solutions.* (An example reference solution is provided on canvas)

- Concisely state the information given and the problem to be solved. Do not copy the problem verbatim; briefly restate the problem in a way that is meaningful to you.
- Provide a neat sketch of the hardware to be analyzed and the location of any system boundaries used for analysis. Indicate next to the boundaries whether the analysis is to be performed as a control mass (CM) or control volume (CV). List particulars of the information given on, or adjacent to, the sketch.
- Show and label any transfers (mass, heat, energy) across the system boundary using arrows to indicate the assumed direction. Your balance equations should be written using the directional convention shown on your sketch.
- List any relevant assumptions that you are making in the analysis: for example, a steady state assumption or neglecting changes in potential energy.
- Sketch the process path on an appropriate thermodynamic diagram (p - v , s - T , p - h , ...). Number each state for reference and clearly denote the nature of any process path (e.g., constant volume). If the exact process path is not known, connect the state points with a dashed line. Carefully label all coordinate axes and pathways with corresponding thermodynamic processes.
- Apply the appropriate balance equations (mass, momentum, energy, entropy) and any equations of state required to complete your analysis. Carry the analysis through to completion *in symbolic form* before substituting numerical values.
- Obtain any property information required and substitute into your analytical expression to obtain the numerical answer. Always carry units when substituting numerical values and check to ensure that they cancel correctly.
- Always assess whether your answer is reasonable (negative pressure, excessively high temperature, decrease in entropy?), and briefly discuss your results.

² <https://communitystandards.stanford.edu/policies-guidance/honor-code>

Format of the problem-set solution: In addition to the solution approach, adhere to the following formatting guideline

- All solutions should be carried out and presented in SI units only.
- Clearly identify your answer by placing a box around it.
- Indicate all problems attempted on top of the front page.
- Staple all sheets together and hand in flat. (No paper clips – this is how pages get lost!)

Grading Policy: Problem grading will be done by use of a ten-point scale. For problems worth more than 10 points, scores will be scaled accordingly.

- **10-9 points:** a very good to excellent solution containing perhaps one small error. Small errors include minor numerical errors.
- **8-7 Points:** a good solution, containing one major error or several small errors. Major errors include improper use of basic principles, bad mathematics, poor use of control-masses or control-volumes, failure to use the solution format presented above or to provide a process diagram.
- **6-5 Points:** A poor solution might contain two major errors or one major error and several small errors.
- **4-1 Points:** A very poor solution represents a minimal attempt at the solution

Request for regarding: If you believe that the grader has made a mistake on your homework, you must write down a description of the error(s) and staple it to the front of the homework. Submit this to Prof. Ihme or the Teaching team *within a week* after you have received the graded homework in class. Petitions that are submitted after one week will not be considered. The same rule applies to midterms and exams.